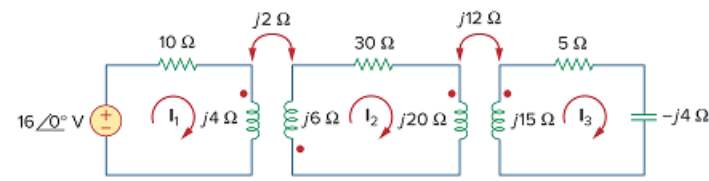


Problem

Find currents I_1 , I_2 , and I_3 in the circuit of Fig. 13.104.

Figure 13.104



Step-by-step solution

Step 1 of 1

For Mesh 1,

$$16 = (10 + j4)I_1 + j2I_2 \rightarrow (1)$$

For Mesh 2,

$$0 = j2I_1 + (30 + j26)I_2 - j12I_3 \rightarrow (2)$$

For Mesh 3,

$$0 = -j12I_2 + (5 + j11)I_3 \rightarrow (3)$$

We may use Matlab to solve (1) to (3) and obtain,

$$I_1 = 1.3736 - j0.5385 = 1.4754 \angle -21.41^\circ \text{ A}$$

$$I_2 = -0.0547 - j0.0549 = 0.0775 \angle -134.85^\circ \text{ A}$$

$$I_3 = -0.0268 - j0.0721 = 0.077 \angle -110.41^\circ \text{ A} .$$